MULTIMEDIA SUPPORT AND MAINTENANCE WEEK13

NAME: TANKISO MASOEBE

1.Personal Backup Audit and Plan

1. Introduction

Data protection is essential for preventing the loss of important personal files such as school projects, photos, and documents. This report evaluates my current data protection practices, identifies weaknesses based on the 3-2-1 backup rule, and proposes an improved, practical backup strategy

2. Current Data Protection Practices (Audit)

At present, my personal data is mainly stored on my laptop, which serves as the primary working device. Some files are also stored on my smartphone. Occasionally, I copy important files to a USB flash drive; however, this process is manual and not done consistently. There is no automated backup system, and cloud storage is either not used or used irregularly.

Summary of Current Setup

- Primary storage: Laptop
- Secondary storage: Smartphone
- Occasional backup: USB flash drive (manual)
- Cloud storage: Not consistently used

3. Weaknesses Identified (3-2-1 Rule Analysis)

The 3-2-1 backup rule recommends having three copies of data, stored on two different types of media, with one copy stored offsite.

The following weaknesses were identified in my current setup:

- Lack of sufficient copies: Most files exist in only one or two locations
- No reliable offsite backup: Absence of consistent cloud storage usage
- Lack of automation: Backups depend on manual effort, increasing the risk of neglect
- Outdated backups: USB backups are irregular and may not contain recent files

- High risk of data loss: If the laptop is damaged, stolen, or crashes, important data could be permanently lost

4. Proposed Backup Plan

Objective

To implement a reliable backup system that follows the 3-2-1 rule and ensures data safety.

Backup Structure

Primary Copy (Working Data):

- Stored on laptop for daily use

Secondary Copy (Local Backup):

- Stored on an external hard drive (1TB or more)
- Automated using Windows File History

Third Copy (Offsite Backup):

- Stored in cloud services such as Google Drive, OneDrive, or Dropbox
- Important folders (Documents, Pictures) synchronized automatically

5. Backup Schedule

Day	Activity
Daily	Automatic cloud synchronization
Wednesday	Verify cloud backup status
Friday	Automatic backup to external hard drive
Sunday	Manual review and file organization

6. Tools and Resources Required

- External hard drive (minimum 1TB capacity)

- Cloud storage service (e.g., Google Drive, OneDrive, or Dropbox)
- Backup software:
    - Windows File History (local backups)
    - Cloud synchronization applications

7. Benefits of the Proposed Plan

- Reduces the risk of permanent data loss
- Protects against hardware failure, theft, or accidental deletion
- Provides multiple recovery options
- Ensures up-to-date backups through automation
- Improves overall organization of files

8. Conclusion

The current data protection approach is insufficient and exposes important files to significant risk. By implementing the proposed 3-2-1 backup strategy, including both local and cloud backups with a structured schedule, data security and reliability will be greatly improved.

2.Data Recovery Simulation Report

1. Introduction

Data loss can occur due to accidental deletion, system failure, or malware. This simulation demonstrates how deleted files can be recovered using data recovery software. The purpose is to evaluate the effectiveness of recovery tools and understand the challenges involved.

2. Methodology

Step 1: Preparation of Test Data

A USB flash drive was used as a non-critical storage device. A folder named **"Test Recovery"** was created, containing the following files:

- 2 Microsoft Word documents
- 2 image files (JPG format)
- 1 PDF document

Step 2: File Deletion

The folder was permanently deleted using **Shift + Delete**, which bypasses the recycle bin. This ensured that the files were not easily recoverable through standard methods.

Step 3: Data Recovery Process

The free recovery tool Recuva was used to attempt file recovery.

Procedure:

1. Installed and launched Recuva
2. Selected the USB drive as the target location
3. Chose "All Files" for recovery
4. Enabled deep scan for better results
5. Started the scan and waited for completion
6. Selected recoverable files and restored them to a different location

3. Results

File Type	Total Files Deleted	Files Recovered	Success Rate
Word	2	2	100%
Images	2	2	100%
PDF	1	1	100%

Overall Success Rate: 100%

All deleted files were successfully recovered. Most files were intact and usable after recovery.

4. Challenges Encountered

- Some files appeared with unclear or missing names
- The deep scan process took a long time
- It was difficult to identify files without previewing them
- Risk of overwriting deleted data if the USB drive is used before recovery

5. Key Observations

- Deleted files are not immediately erased; they remain recoverable until overwritten
- Using the storage device after deletion reduces recovery chances
- Deep scan significantly improves recovery success
- File recovery tools are effective for recently deleted files

6. Conclusion

The simulation demonstrated that data recovery software can successfully restore permanently deleted files, especially if recovery is attempted immediately. However, delays or continued use of the storage device may reduce success rates. This highlights the importance of both proper backup strategies and cautious handling of deleted data.

7. Recommendation

To minimize data loss risks:

- Always maintain backups (following the 3-2-1 rule)
- Avoid using a device after accidental deletion
- Use recovery tools immediately when data is lost
- Consider using reliable recovery software when needed